

SUGAR LAND, TX, USA

WMO: 722543

Lat: 29.622N

Lon: 95.657W

Elev: 84

StdP: 14.65

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12977

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.4	(c) 34.6	(d) 17.7	(e) 13.5	(f) 42.9	(g) 21.6	(h) 16.3	(i) 45.3	(j) 24.3	(k) 59.8	(l) 21.9	(m) 61.5	(n) 5.8	(o) 0	(p) 0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 18.4	(c) 97.3	(d) 76.7	(e) 95.2	(f) 77.0	(g) 93.4	(h) 77.1	(i) 80.5	(j) 88.6	(k) 79.9	(l) 88.2	(m) 79.2	(n) 87.6	(o) 8.1	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	150.7	83.2	77.9	145.2	83.0	77.1	141.5	82.7	44.1	88.5	43.2	87.8	42.6	87.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.9	(b) 18.9	(c) 17.2	(e) 25.6	(f) 100.4	(g) 3.8	(h) 2.8	(i) 22.8	(j) 102.4	(k) 20.6	(l) 104.0	(m) 18.5	(n) 105.6	(o) 15.7	(p) 107.6		
(4) 20.9	(5) 18.9	(6) 17.2	(7) 25.6	(8) 100.4	(9) 3.8	(10) 2.8	(11) 22.8	(12) 102.4	(13) 20.6	(14) 104.0	(15) 18.5	(16) 105.6	(17) 15.7	(18) 107.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.5	53.5	57.9	64.0	70.4	77.2	82.6	83.9	84.4	80.4	72.1	62.8	56.0
	DBStd	12.94	9.38	9.84	8.25	6.39	5.02	2.98	2.64	2.78	4.01	7.28	9.11	9.52
	HDD50	170	69	37	7	0	0	0	0	0	0	0	11	46
	HDD65	1237	372	235	121	26	2	0	0	0	0	24	147	310
	CDD50	7647	178	257	439	611	844	978	1051	1067	911	686	394	231
	CDD65	3240	16	35	88	186	381	528	586	602	461	245	81	31
	CDH74	32511	56	133	430	1230	3464	5945	6822	7097	4552	2159	498	125
	CDH80	13774	1	8	51	252	1238	2746	3248	3516	1919	707	82	6
(14) Wind	WSAvg	7.7	8.3	9.0	9.2	9.5	9.0	7.3	6.2	5.9	5.8	6.7	7.4	8.0
(15) Precipitation	PrecAvg	49.9	3.5	2.8	3.6	4.2	4.8	4.7	4.0	4.3	4.4	5.0	4.2	3.7
	PrecMax	69.3	9.9	9.2	8.6	10.8	12.4	13.4	13.3	9.3	14.1	12.8	14.3	9.4
	PrecMin	17.5	0.1	0.3	0.5	0.0	0.0	0.2	0.7	0.4	0.9	0.0	0.2	0.7
	PrecStd	13.0	2.4	2.2	2.2	3.2	3.3	3.2	3.3	2.5	3.0	4.0	3.2	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.7	81.3	84.3	88.4	93.5	98.4	99.0	100.5	97.2	91.4	86.0	80.9
		MCWB	67.3	68.6	69.3	72.9	74.8	75.7	76.8	76.6	75.5	76.1	73.0	70.4
	2%	DB	75.2	78.0	81.2	84.9	90.8	95.3	96.7	97.6	93.8	89.3	82.2	77.5
		MCWB	66.0	67.3	69.2	71.5	74.9	76.2	76.8	77.3	76.3	75.5	71.2	69.2
	5%	DB	72.4	75.0	78.9	82.7	88.9	93.1	94.4	95.3	91.4	87.0	79.6	74.6
		MCWB	65.3	66.8	67.8	70.9	74.4	76.7	77.3	77.7	75.9	73.7	69.6	68.5
	10%	DB	69.7	72.5	76.0	80.9	86.8	91.1	92.5	93.2	89.7	84.1	76.7	72.1
		MCWB	63.8	66.1	66.8	69.7	74.0	76.7	77.4	77.7	75.8	71.8	68.8	67.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.2	71.9	73.1	76.5	79.0	81.3	81.0	81.8	80.5	79.2	76.1	73.6
		MCDB	74.6	76.9	79.2	84.3	86.8	88.6	89.2	89.9	88.0	85.7	81.4	76.7
	2%	WB	69.2	70.4	71.6	74.6	77.5	80.0	80.1	80.6	79.6	78.1	74.2	71.7
		MCDB	72.6	74.8	77.2	81.2	85.3	87.8	88.7	89.1	87.3	84.7	78.9	75.3
	5%	WB	67.4	69.1	70.6	73.5	76.7	79.0	79.4	80.0	78.7	76.9	72.5	70.2
		MCDB	71.3	73.2	75.8	79.3	84.3	87.1	88.1	88.6	86.3	83.4	77.2	73.9
	10%	WB	64.9	67.4	69.4	72.5	75.9	78.2	78.7	79.2	77.9	75.6	70.7	67.7
		MCDB	68.4	71.4	74.2	78.0	83.2	86.6	87.3	87.9	85.4	81.3	75.4	71.3
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	18.3	18.6	18.5	17.8	17.4	17.3	18.4	17.6	20.4	20.3	19.2
		MCDBR	21.5	19.2	18.7	18.6	19.0	19.8	20.2	21.0	20.0	19.5	20.1	20.0
		MCWBR	13.9	11.5	9.3	8.4	6.2	5.3	5.2	5.3	6.3	7.6	10.9	13.2
	5% WB	MCDBR	17.9	15.9	15.4	14.8	15.5	16.2	17.0	18.1	16.6	16.3	16.8	17.8
		MCWBR	13.5	11.6	9.4	8.2	6.1	5.8	5.6	5.7	5.9	8.1	11.3	14.0
(40) Clear-Sky Solar Irradiance	taub		0.344	0.360	0.376	0.438	0.484	0.468	0.496	0.485	0.455	0.381	0.375	0.349
	taud		2.482	2.450	2.395	2.223	2.129	2.228	2.169	2.201	2.253	2.447	2.413	2.473
	Ebn at Noon		283	287	289	273	259	262	254	257	261	276	268	274
	Edh at Noon		29	33	37	45	50	45	48	46	42	32	31	28
(44) All-Sky Solar Radiation	RadAvg		911	1029	1401	1673	1859	1941	1907	1829	1578	1372	1033	810
	RadStd		89	180	120	124	136	156	181	145	165	163	107	99

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.39	N/A	N/A	+1.37	N/A	N/A	N/A	+423	+304
(47) Regional (33 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	+200	+175

Nomenclature: See separate page